

A democratic-stress threshold for early warning of democratic erosion and collapse

Thomas Hedlund Leijon

FieldShift Research, Stockholm, Sweden

Correspondence: thomas.hedlundleijon@gmail.com

Abstract

Democratic erosion often appears gradual until institutions cross into elevated vulnerability. Here I introduce an early-warning framework derived from a geometric model of democratic stability. Three institutional pillars—rule of law, horizontal accountability, and democratic norms—are represented as vectors at 120° angles. Their triangle area defines institutional capacity. A critical stress threshold $S_2 = \sqrt{3}/4 \approx 0.4330$ emerges as the bifurcation point where symmetric recovery ceases to be the unique stable equilibrium. Using V-Dem v16 (199 countries, 1790–2025), this pre-registered threshold identifies a robust high-vulnerability regime (odds ratio 20.56 [11.94, 35.43]). Critical slowing down indicators distinguish pre-collapse from stable high-stress observations (variance: $\beta = 0.00126$, $p = 2.0 \times 10^{-66}$, Cohen's $d = 0.685$; autocorrelation: $\beta = 0.0814$, $p = 3.1 \times 10^{-13}$, $d = 0.289$), while the naive prediction—uniformly rising indicators above S_2 —is decisively falsified ($p = 1.4 \times 10^{-127}$). An asymmetry hypothesis $\mathcal{K} = \sqrt{3}$ for recovery-erosion imbalance is tested against five competing values across three episode definitions and two magnitude measures. Across all six specifications, $\sqrt{3}$ is the only value consistently within bootstrap confidence intervals; four alternatives, including the symmetry null, are excluded. Point estimates range from 1.55 to 1.96. The framework provides a transparent, falsifiable early-warning classifier for democratic erosion.

1. Introduction

Democratic breakdown is usually studied after the fact, yet the policy and scientific need for early detection of institutional deterioration is acute. Existing democracy datasets provide rich measurement, but the translation of such measurement into transparent, theoretically motivated early-warning classification remains underdeveloped. A useful early-warning framework must be interpretable, pre-specified, robust to alternative outcome definitions, and explicit about its inferential limits.

This article develops and validates a democratic-stress framework derived from a geometric model of institutional stability. The central claim is that three foundational institutional capacities—rule of law (R), horizontal accountability (M), and democratic norms (N)—are equally fundamental to democratic equilibrium. When represented as vectors in the plane with equal 120° separation, their joint capacity can be measured as the area of the triangle they span.

From this representation, three constants emerge. A geometrically motivated early-warning level $S_1 = S_2/2 = \sqrt{3}/8$ marks the midpoint between full stability and the critical threshold. A critical threshold $S_2 = \sqrt{3}/4$ marks the bifurcation point where symmetric recovery ceases to be the unique stable equilibrium—derived deductively from the area of the unit equilateral triangle. An asymmetry hypothesis $\mathcal{K} = \sqrt{3}$ conjectures that the magnitude of recovery-erosion asymmetry is governed by the same geometric constant. While S_2 follows deductively from the geometric representation, $\mathcal{K}$ is a conjectural extension; both are empirically testable and falsifiable.

The contribution is threefold. First, the article introduces a transparent geometric model providing theoretical rationale for pre-specified thresholds. Second, it validates the S_2 threshold against democratic collapse and demonstrates a two-stage early-warning logic: S_2 defines who is at risk; critical slowing down signals indicate when collapse is imminent. Third, it tests the asymmetry hypothesis against competing geometric values using two complementary magnitude operationalisations and three episode definitions.

2. A geometric theory of democratic stability

2.1 The three-pillar representation

We represent the institutional foundations of democracy as three vectors in $\mathbb{R}^2$ sharing a common origin O . Each vector encodes the strength of one institutional pillar on the unit interval $[0,1]$:

$$\begin{aligned} \mathbf{r}_R &= (R, 0) && \text{Rule of law} \\ \mathbf{r}_M &= (-M/2, (\sqrt{3}/2)M) && \text{Horizontal accountability} \\ \mathbf{r}_N &= (-N/2, -(\sqrt{3}/2)N) && \text{Democratic norms} \end{aligned} \quad (1)$$

The equal 120° angular separation ($360^\circ/3$) encodes the assumption that the three pillars are equally fundamental in sustaining democratic equilibrium. The endpoints of the three vectors define a triangle whose area provides a scalar measure of total institutional capacity.

Area derivation. Define side vectors $\mathbf{u} = \mathbf{r}_M - \mathbf{r}_R$ and $\mathbf{v} = \mathbf{r}_N - \mathbf{r}_R$. The area is half the absolute value of the two-dimensional cross product:

$$A(R,M,N) = (1/2)|\mathbf{u} \times \mathbf{v}| = (\sqrt{3}/4)(RM + RN + MN) \quad (2)$$

Full derivation is provided in Supplementary Information, Section S1.

Maximum area. When $R = M = N = 1$: $A_{\max} = 3\sqrt{3}/4 \approx 1.299$.

Normalised democratic stability:

$$\Phi(R,M,N) = A/A_{\max} = (RM + RN + MN)/3 \in [0,1] \quad (3)$$

Democratic stress:

$$S = 1 - \Phi = 1 - (RM + RN + MN)/3 \in [0,1] \quad (4)$$

A fully healthy democracy has $S = 0$; complete institutional disintegration corresponds to $S = 1$.

2.2 Derivation of the critical thresholds

The early-warning level S_1 . A geometrically natural candidate for an early-warning indicator is the midpoint on the stress scale between full stability and the critical threshold:

$$S_1 = S_2/2 = \sqrt{3}/8 \approx 0.2165 \quad (5)$$

Unlike S_2 , S_1 is a conventional level whose utility is evaluated empirically.

The critical threshold S_2 . The area of an equilateral triangle with side length 1 is $\sqrt{3}/4$. In normalised stress:

$$S_2 = \sqrt{3}/4 \approx 0.4330 \quad (6)$$

At this stress, normalised stability $\Phi_c = 1 - \sqrt{3}/4 \approx 0.567$. No empirical fitting is involved: S_2 follows deductively from the geometry.

The logical structure is as follows: S_2 is defined geometrically. The bifurcation analysis in Section 2.3 demonstrates that a pitchfork bifurcation can be placed at this geometrically defined value—establishing consistency, not derivation.

2.3 Bifurcation at the geometric threshold

To assess whether the geometrically defined threshold S_2 is compatible with a dynamical mechanism producing accelerated erosion, we construct a minimal symmetry-preserving model. Let stability Φ evolve in a potential:

$$U(\Phi; S) = (1/2)k(S)(\Phi - \Phi_c)^2 + (\lambda/4)(\Phi - \Phi_c)^4, \quad \lambda > 0 \quad (7)$$

with control parameter $k(S) = \alpha(S_2 - S)$, $\alpha > 0$.

Equilibria. Setting $\partial U / \partial \Phi = 0$:

$$(\Phi - \Phi_c)[k(S) + \lambda(\Phi - \Phi_c)^2] = 0 \quad (8)$$

Solutions are $\Phi = \Phi_c$, which always exists; and, when $k(S) < 0$, $\Phi = \Phi_c \pm \sqrt{-k(S)/\lambda}$.

Stability. The second derivative is $\partial^2 U / \partial \Phi^2 = k(S) + 3\lambda(\Phi - \Phi_c)^2$. At $\Phi = \Phi_c$, this reduces to $k(S)$. Thus $S < S_2$ implies $k(S) > 0$ and Φ_c is stable; $S = S_2$ implies $k(S) = 0$ and marginal stability; $S > S_2$ implies $k(S) < 0$ and Φ_c is unstable. At the non-zero branches for $S > S_2$, $\partial^2 U / \partial \Phi^2 = -2k(S) > 0$, so the branches are stable.

The system undergoes a supercritical pitchfork bifurcation at $S = S_2$. Below the threshold there is a unique stable equilibrium. At the threshold the recovery rate vanishes, producing critical slowing down. Above the threshold the system becomes bistable, and noise can drive collapse onto the lower branch.

The model is phenomenological: it demonstrates consistency with a bifurcation mechanism, not unique derivation. The empirical validation tests consequences of the model, not the model itself.

2.4 The asymmetry hypothesis: $\mathcal{H} = \sqrt{3}$

A well-documented empirical observation is that democratic recovery requires substantially greater institutional effort than the erosion preceding it—democracies are easier to break than to fix. The geometric framework suggests a natural conjecture: if this asymmetry reflects the same structural imbalance producing the critical threshold at $S_2 = \sqrt{3}/4$, then the recovery-to-erosion magnitude ratio might approximate $\sqrt{3}$.

This is a hypothesis, not a deduction. Define erosion magnitude M^- and recovery magnitude M^+ for episode pairs. The asymmetry constant is:

$$\mathcal{H} = \text{mean}(M^+) / \text{mean}(M^-) \quad (9)$$

Symmetry implies $\mathcal{H} = 1$. The geometric conjecture is $\mathcal{H} = \sqrt{3} \approx 1.7321$.

Falsifiability. We test five geometrically motivated values against bootstrap confidence intervals: 1 (symmetry null), $\sqrt{3}/2$, $\sqrt{3}$, $2\sqrt{3}$, and $3\sqrt{3}$. If $\sqrt{3}$ is uniquely supported, only it should fall within the empirical confidence interval. If multiple values do, the hypothesis lacks specificity. If none do, the geometric approach to asymmetry is falsified.

Two magnitude operationalisations. We employ two complementary measures. Net change, $|\text{end} - \text{start}|$, captures total capacity loss or gain—the closest operationalisation to “how much did democracy fall, and how much must it rise?” Sum of absolute annual deltas additionally captures within-episode

volatility—“total institutional turbulence.” Both are reported; neither is privileged. This pre-committed dual reporting eliminates specification searching.

2.5 Falsifiable predictions

The geometric model generates three testable predictions:

P1 (Classification): Country-years with $S \geq S_2$ exhibit substantially elevated collapse risk, robust across alternative specifications.

P2 (Critical slowing down): The naive prediction—uniformly rising variance and autocorrelation above S_2 —is likely false: post-bifurcation systems may lock into stable low-capacity equilibria with dampened fluctuations. The refined prediction is that within the $S \geq S_2$ population, pre-collapse observations exhibit elevated variance and autocorrelation relative to stable high-stress observations.

P3 (Asymmetry): $\mathcal{H}_0$ is consistent with $\sqrt{3}$ across episode definitions and magnitude measures; $\sqrt{3}$ is uniquely supported relative to four competing geometric values.

2.6 On the distinction from numerology

A legitimate concern is whether the appearance of $\sqrt{3}$ constitutes scientific discovery or mere numerology. The distinction lies in falsifiability. The value $\sqrt{3}/4$ is not asserted to be special by fiat; it follows from three explicit, challengeable assumptions: equal fundamentality of three pillars, area representation of capacity, and symmetry-preserving bifurcation structure. Relaxing any assumption changes the threshold. The framework makes testable claims that are in principle falsifiable. The appearance of $\sqrt{3}$ is a consequence of three-component symmetry—analogue to π in circular problems—not an asserted magical property.

3. Data and methods

3.1 Data

The analysis uses the V-Dem Country-Year Full+Others dataset, version 16, published in March 2026 by the Varieties of Democracy Institute. The analytic panel contains 25,509 country-year observations covering 199 countries from 1790 to 2025. The primary democracy measure is the V-Dem Liberal Democracy Index ($D_{it} \in [0,1]$).

Democratic stress is defined as $S_{it} = 1 - D_{it}$. The focal threshold $S_2 = \sqrt{3}/4 \approx 0.4330$ was pre-specified before any model fitting on V-Dem data. The pre-specification document and code commit hashes are archived with the replication package.

3.2 Collapse definition

The primary collapse outcome is an annual decline in democratic performance of at least 0.05: $\text{collapse}_{it} = 1[D_{it} - D_{i,t-1} \leq -0.05]$. Forward-collapse outcomes for horizon h are defined as $\text{collapse}_{it}^{(h)} = 1[\max(\text{collapse}_{i,t+1}, \dots, \text{collapse}_{i,t+h}) = 1]$ for $h \in \{1,2,3,5\}$. Alternative thresholds of 0.03 and 0.07 are reported as robustness checks. External outcomes use V-Dem Regimes of the World coding.

3.3 Early-warning indicators

For each country with a time series of democratic stress, we compute rolling-window indicators of critical slowing down with primary window length $w = 10$ years. Within each window, we detrend the stress

series linearly and compute the variance and lag-1 autocorrelation of the residuals. Robustness to window lengths $w \in \{5, 8, 12, 15\}$ is evaluated.

3.4 Model specifications

Classification models. Baseline models estimate collapse risk as a function of lagged S_2 exposure via logistic regression. Robustness specifications include country-clustered standard errors, year fixed effects, country-and-year fixed effects, alternative democracy measures (Polyarchy), percentile-mapped composite stress, leave-one-region-out and leave-one-period-out analyses, and temporal holdout validation (training: 1790–2009; holdout: 2010–2025).

Early-warning tests. Three statistical tests evaluate the critical slowing down predictions:

Test 1 (Above vs. below S_2): Linear models of variance and AR(1) on S_2 exposure with country-clustered standard errors. This test evaluates the naive prediction of uniformly rising critical slowing down indicators. Within-country paired comparisons (countries with at least five observations both above and below S_2) corroborate the pooled results.

Test 2 (Pre-collapse vs. stable high-stress): Within the $S \geq S_2$ population, linear models of variance and AR(1) on pre-collapse status with country-clustered standard errors. Pre-collapse is defined as $S_{it} \geq S_2$ and $\text{collapse}_{it} = 0$. The reference group is $S_{it} \geq S_2$ and $\text{collapse}_{it} = 0$.

Test 3 (Temporal trend before collapse): For each collapse event with at least four pre-collapse observations, linear trend models of variance and AR(1) on time to collapse. One-sample t-tests assess whether mean slopes exceed zero.

Placebo test. Collapse timing is randomised within countries while preserving within-country event rates, using 1,000 permutations. The observed mean trend slope is compared to the permutation distribution; exact one-sided p-values are reported.

Asymmetry estimation. Erosion and recovery episodes are identified using three alternative definitions of increasing stringency: Liberal (≥ 2 years, no minimum magnitude), Standard (≥ 5 years, total magnitude ≥ 0.10 ; approximating Episodes of Regime Transformation criteria), and Strict (≥ 8 years, ≥ 0.20). We apply both magnitude thresholds (abs-sum and net) to ensure only episodes meeting both criteria qualify, providing a stricter selection. This reduces the Standard-definition sample from 45 to 35 episode pairs.

Two magnitude measures are computed. Net change is $|\text{end value} - \text{start value}|$. Sum of absolute annual deltas is $\sum |\Delta D_t|$ within the episode. For each definition and magnitude measure, erosion-recovery episode pairs are identified within countries. $\mathcal{M}_{\text{net}} = \text{mean}(M^+)/\text{mean}(M^-)$, with bias-corrected and accelerated bootstrap 95% confidence intervals based on 10,000 resamples. Five geometrically motivated values are tested against each interval.

4. Results

4.1 Classification: S_2 identifies a high-vulnerability regime

Across the analytic panel of 25,509 country-year observations, the S_2 regime is strongly associated with the primary collapse outcome. In the baseline logistic specification, the estimated odds ratio is 20.56 [11.94, 35.43] (AUC 0.735, PR lift 2.10 $\times$). The effect remains positive with country-clustered uncertainty estimates, year fixed effects, and country-by-year fixed-effects specifications.

The signal is robust across alternative collapse thresholds: for annual declines of 0.03, 0.05, and 0.07, odds ratios range from 14.90 to 34.01, with stronger effects under stricter collapse definitions. The effect

persists across forward-looking horizons: lagged S_2 exposure predicts collapse within one, two, three, and five years (OR: 13.19 to 15.14).

Replication using Polyarchy stress yields OR 8.82 [5.24, 14.84] (AUC 0.765). A percentile-mapped composite stress specification yields OR 7.86 [5.06, 12.20] (AUC 0.781). External validation supports the same interpretation: S_2 exposure is associated with V-Dem Regimes of the World downgrade outcomes and democracy-to-autocracy transitions.

Leave-one-region-out analyses yield positive S_2 effects across all regional exclusions (OR: 14.93 to 36.73). Leave-one-period-out analyses remain positive after excluding the pre-1945, 1945–1989, 1990–2009, and 2010–2025 periods.

4.2 Temporal holdout validation

In the 2010–2025 holdout period, the transparent model for three-year forward collapse yields AUC 0.603, PR lift 1.62 $\times$, Brier score 0.0735, and expected calibration error 0.0387. The top 10% of risk-ranked observations capture 19.8% of future three-year collapses, the top 20% capture 37.0%, and the top 30% capture 48.0%. This pattern is interpreted under rare-event conditions, where risk enrichment rather than maximised discrimination is the primary evaluation criterion.

4.3 Critical slowing down within the vulnerable population

Test 1: Above vs. below S_2 . The naive critical slowing down prediction—uniformly rising variance and autocorrelation as stress increases toward the bifurcation point—is decisively falsified. The general population of high-stress ($S \geq S_2$) country-years exhibits lower average variance ($\bar{V} = 0.00051$ vs. 0.00199) and lower average autocorrelation (AR(1) mean = 0.208 vs. 0.259) than country-years below the threshold. The difference is highly significant for both variance ($\beta = -0.00147$, $p = 1.4 \times 10^{-127}$ with country-clustered standard errors) and autocorrelation ($\beta = -0.0507$, $p = 3.1 \times 10^{-21}$). Within-country paired comparisons ($n = 56$ countries) corroborate the negative pooled effect: the mean within-country difference is negative for both variance ($t = -1.87$, $p = 0.97$ against the one-sided hypothesis) and AR(1) ($t = -2.10$, $p = 0.98$).

This falsification is informative. It demonstrates that the $S \geq S_2$ regime is not uniformly approaching a transition. Rather, many high-stress systems appear locked into stable low-capacity equilibria with dampened fluctuations—consistent with bistable dynamics. This motivates Test 2.

Test 2: Pre-collapse vs. stable high-stress. The refined prediction is strongly confirmed. Within the $S \geq S_2$ population, pre-collapse observations ($n = 660$) exhibit significantly elevated variance ($\beta = 0.00126$, $p = 2.0 \times 10^{-66}$; Cohen's $d = 0.685$) and elevated autocorrelation ($\beta = 0.0814$, $p = 3.1 \times 10^{-13}$; Cohen's $d = 0.289$) relative to stable high-stress observations ($n = 18,207$). The stable group shows mean variance 0.00044 and mean AR(1) 0.201; the pre-collapse group shows mean variance 0.00170 and mean AR(1) 0.283.

Test 3: Temporal trend before collapse. Among 291 collapse events with sufficient pre-collapse data, variance shows a positive but non-significant trend over the five years preceding collapse (mean $\beta = 6.71 \times 10^{-5}$, $p = 0.086$), with 60.1% of events exhibiting positive variance trends. Autocorrelation shows a weak negative trend (mean $\beta = -4.97 \times 10^{-3}$, $p = 0.828$), with 48.8% positive. The absence of a robust monotonic trend suggests that transitions may be relatively abrupt rather than gradual—consistent with noise-driven barrier crossing in bistable systems.

Table 1. Critical slowing down results.

Test	Indicator	Coefficient	p-value	Cohen's d
Test 1: above vs. below S_2	Variance	-1.47×10^{-3}	1.4×10^{-127}	—
Test 1: above vs. below S_2	AR(1)	-5.07×10^{-2}	3.1×10^{-21}	—
Test 2: pre-collapse vs. stable	Variance	$+1.26 \times 10^{-3}$	2.0×10^{-66}	0.685
Test 2: pre-collapse vs. stable	AR(1)	$+8.14 \times 10^{-2}$	3.1×10^{-13}	0.289
Test 3: trend before collapse	Variance trend	$+6.71 \times 10^{-5}$	0.086	—
Test 3: trend before collapse	AR(1) trend	-4.97×10^{-3}	0.828	—

4.4 Placebo diagnostics

Permuting collapse timing within countries (1,000 draws) generates a null distribution for the pre-collapse trend estimates. The observed mean variance trend ($\beta = 6.71 \times 10^{-5}$) exceeds 90.7% of the placebo distribution ($p = 0.093$). The observed mean autocorrelation trend ($\beta = -4.97 \times 10^{-3}$) exceeds only 8.5% of the placebo distribution ($p = 0.915$). The classification S_2 signal itself exceeds the 99th percentile of the placebo distribution (OR = 3.18 vs. observed OR = 20.56), confirming that the threshold signal is not an artefact of random timing.

4.5 Institutional mechanisms

Matched event study comparing collapse cases with high-stress non-collapse controls shows that the collapse pattern is not reducible to high baseline stress alone. Collapse cases diverge sharply at event time $t = 0$, while matched controls remain comparatively stable. Mechanism-lag difference-in-differences analyses reveal pre-onset deterioration among collapse cases relative to controls in all ten institutional mechanism scores examined: vertical accountability, horizontal accountability, freedom of expression, freedom of association, civil liberties, executive constraints, legislative constraints, judicial constraints, rule of law, and electoral democracy.

4.6 The asymmetry hypothesis: empirical evaluation

The geometric conjecture predicts that democratic recovery requires approximately $\sqrt{3} \approx 1.7321$ times the cumulative institutional effort of the preceding erosion. This hypothesis is tested independently of the threshold validations in Sections 4.1–4.5.

Table 2. Recovery-erosion asymmetry across three episode definitions and two magnitude measures, V-Dem v16.

Definition	Magnitude	N pairs	N countries	Mean M ⁻	Mean M ⁺	$\mathcal{H}_c \square \square$	95% BCa CI
Liberal (≥ 2 yr)	Abs sum	1,305	195	0.038	0.064	1.682	[1.510, 1.881]
Liberal (≥ 2 yr)	Net	1,305	195	0.026	0.052	1.962	[1.709, 2.258]
Standard (≥ 5 yr, ≥ 0.10)	Abs sum	35	32	0.294	0.456	1.551	[1.359, 1.778]
Standard (≥ 5 yr, ≥ 0.10)	Net	35	32	0.271	0.427	1.576	[1.366, 1.820]
Strict (≥ 8 yr, ≥ 0.20)	Abs sum	10	10	0.368	0.636	1.727	[1.462, 2.076]
Strict (≥ 8 yr, ≥ 0.20)	Net	10	10	0.358	0.611	1.706	[1.429, 2.060]

Table 3. Tests of geometrically motivated hypotheses against bootstrap confidence intervals.

Hypothesis d %	Decimal	Lib (abs)	Lib (net)	Std (abs)	Std (net)	Str (abs)	Str (net)
1 (symmetry null)	1.000	excluded	excluded	excluded	excluded	excluded	excluded
$\sqrt{3}/2$	0.866	excluded	excluded	excluded	excluded	excluded	excluded
$\sqrt{3}$	1.732	within CI	within CI	within CI	within CI	within CI	within CI
$2\sqrt{3}$	3.464	excluded	excluded	excluded	excluded	excluded	excluded
$3\sqrt{3}$	5.196	excluded	excluded	excluded	excluded	excluded	excluded

Interpretation. Three findings warrant emphasis. First, $\mathcal{H}_c \square \square$ lies systematically above unity across all six specifications—democracies are consistently harder to rebuild than to erode. Second, $\sqrt{3}$ is the only value within all six bootstrap confidence intervals. Four competing values—including the symmetry null ($\mathcal{H} = 1$)—are excluded in every specification. Third, point estimates range from 1.55 (Standard abs-sum) to 1.96 (Liberal net). The Standard definition, which best matches ERT literature criteria, yields the most conservative estimates (1.55–1.58). The Liberal net-change estimate (1.96) lies above $\sqrt{3}$, illustrating that the operationalisation of episode magnitude can place estimates on either side of the conjectured value. This bidirectional variation strengthens the interpretation that $\sqrt{3}$ is a central reference point rather than a systematically biased estimate. The systematic difference between abs-sum and net measures for the Liberal definition—net yielding substantially higher values—suggests that brief reversals during directional episodes are more common in erosion than in recovery, itself an observation meriting further investigation.

The strict definition, with only 10 episode pairs, yields wide confidence intervals; its inclusion serves robustness documentation rather than precise estimation. The stricter dual-threshold filtering reduces sample sizes for the Standard and Strict definitions; results for these definitions should be interpreted with appropriate caution regarding precision.

5. Discussion

The results support a probabilistic and structural interpretation of democratic stress thresholds. A pre-specified boundary, $S_2 = \sqrt{3}/4$, identifies a robust high-vulnerability regime for democratic collapse and regime downgrade.

A two-stage early-warning logic. The findings refine the threshold's interpretation in a manner consistent with the bifurcation model. Below S_2 , democratic systems are structurally stable; critical slowing down signals are absent because there is no bifurcation to approach. Above S_2 , the system enters a structurally vulnerable regime where the potential landscape has bifurcated. Within this high-risk population, elevated variance and autocorrelation distinguish country-years that subsequently collapse from those that remain

resilient—with effect sizes that are unusually large for observational political science (Cohen’s $d = 0.685$ for variance).

This two-stage logic has direct practical implications. Democracy monitors should first identify countries with stress exceeding S_2 —these constitute the at-risk population. Within this population, rising variance and autocorrelation in the stress time series provide the near-term warning. The naive prediction—uniformly rising CSD indicators above S_2 —is falsified, demonstrating that the post-bifurcation regime is heterogeneous: many high-stress systems are stably locked into low-capacity equilibria, and only a subset exhibit the dynamical signature of approaching collapse.

The asymmetry hypothesis. $\mathcal{K} = \sqrt{3}$ receives qualified empirical support. Across three episode definitions and two magnitude measures, $\sqrt{3}$ is the only geometrically motivated value consistently within all six bootstrap confidence intervals, while four alternatives—including the symmetry null—are excluded in every specification. This survival-against-alternatives constitutes the strongest evidence for the geometric conjecture.

However, the empirical point estimates lie between 1.55 and 1.96, with the ERT-compatible Standard definition yielding the most conservative values (1.55–1.58). $\sqrt{3}$ is empirically consistent but not uniquely confirmed: the evidence favours $\sqrt{3}$ over the tested alternatives, but the residual gap between point estimates and the conjectured value—and the sensitivity to magnitude operationalisation—requires prospective validation. Whether this gap reflects systematic measurement bias, genuine deviation from the geometric prediction, or the inherent noisiness of episode identification will be resolved by replication against future V-Dem releases.

The conjecture’s survival of falsification across multiple specifications—while competing values are excluded—places it in the category of a hypothesis that has passed severe tests, rather than a law-like constant. It remains theoretically motivated, empirically tested, and worthy of independent derivation; it does not carry the deductive weight of $S_2 = \sqrt{3}/4$.

Status of the three constants. $S_1 = \sqrt{3}/8$ is a conventional early-warning level—the midpoint on the stress scale between full stability and the critical threshold. Its utility is empirical. $S_2 = \sqrt{3}/4$ is deductively derived from the geometric definition of the unit equilateral triangle; the bifurcation analysis demonstrates consistency with a plausible dynamical mechanism, not unique derivation. $\mathcal{K} = \sqrt{3}$ is a geometrically motivated hypothesis—a conjectural extension of the same structural logic that produced S_2 , empirically tested against competing values, but awaiting independent theoretical derivation.

Relation to existing frameworks. This framework is complementary to the Regimes of the World classification and the third-wave autocratization literature. Where Regimes of the World identifies categorical regime status, the present framework provides a continuous structural vulnerability indicator operating within regime categories. S_2 exposure can be elevated within an electoral democracy without yet having crossed into electoral autocracy; correspondingly, regime transitions often occur after sustained S_2 exposure.

Relation to critical transitions research. The geometric model connects democratic collapse to the established framework of critical transitions in complex systems. The identification of a bifurcation point with associated early-warning signals links this work to foundational contributions on ecological, climatic, and financial tipping points. The novel contribution is demonstrating that such dynamics can be detected in political institutions, and that the critical threshold can be derived from first principles rather than estimated post hoc.

On falsifiability. Three predictions were specified before analysis. Prediction 1 (classification) is confirmed. Prediction 2 (critical slowing down) is partially confirmed: the naive version is falsified, while

the refined version—CSD signals discriminate pre-collapse from stable states—is strongly supported. The falsification of the naive prediction is itself informative: it reveals that the post-bifurcation regime is heterogeneous, not uniformly approaching transition. Prediction 3 (asymmetry) is consistent with data but not uniquely confirmed: $\sqrt{3}$ survives while competing values are excluded, but point estimates vary across operationalisations. This mixed pattern—confirmation, falsification, and qualified support—is characteristic of a progressive research programme.

Limitations. The findings should be interpreted within five boundaries. First, the geometric model is phenomenological: it demonstrates consistency with a bifurcation mechanism but does not claim to be the unique data-generating process. Second, temporal holdout performance is strongest as risk enrichment rather than maximised discrimination, consistent with rare-event prediction challenges. Third, the mechanism diagnostics are observational; they assess temporal and institutional plausibility rather than experimental causal effects. Fourth, the asymmetry estimates are sensitive to episode definition and magnitude operationalisation, as documented in Table 2. Fifth, prospective validation against future V-Dem releases and external forecasting benchmarks is required before definitive comparative claims. The pre-registered threshold structure and open replication package enable such prospective validation.

Data availability

The analysis uses the V-Dem Country-Year Full+Others dataset, version 16, publicly available from the Varieties of Democracy Institute at <https://v-dem.net/data/>. The replication package—containing preprocessing scripts, derived analysis dataset, validation scripts, and all figure and table outputs—is archived at Zenodo under DOI [to be inserted before submission].

Code availability

The replication package contains all scripts required to reproduce the analysis, together with the exact derived analysis dataset used for the reported results. All analyses were performed in R version 4.5.3.

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